

Developing Action-Oriented Systems for Manual-Computational Craft Workflows

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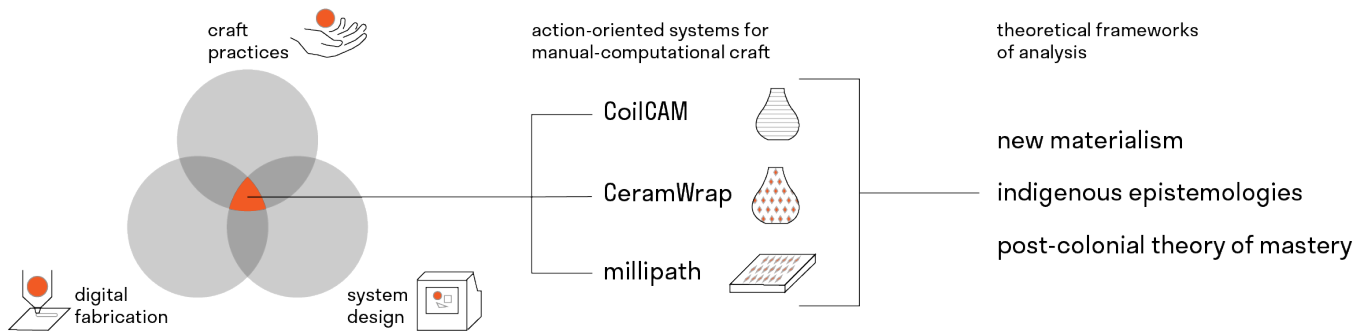


Figure 1: This dissertation proposes works that combine craft practices, digital fabrication, and system design. It presents three action-oriented systems, *CoilCAM*, *CeramWrap*, and *millipath*, which support manual-computational workflows in clay 3D printing, ceramics ornamentation, and CNC milling respectively. Future work will analyze these systems and the artifacts produced with them from the perspective of three theoretical frameworks that can expand HCI system research.

ABSTRACT

Manual crafts typically involve action-oriented workflows, wherein the craftsperson performs a series of repeated actions to create beautiful intricate artifacts. Despite working with similar materials, digital fabrication workflows often implement methods that differ fundamentally from manual ones. In my dissertation work, I aim to develop action-oriented digital fabrication systems that implement abstractions derived from craft domain-expert knowledge. I theorize that action-oriented systems have the potential to (1) support the integration of computational tools with manual practices, (2) enable complex design tasks that would be challenging to achieve through other means, and (3) leverage the unique properties of materials through fine control over machine toolpath. To investigate this theory, I developed three digital fabrication systems, which I evaluated by producing various artifacts. Additionally, I propose analyzing my work and HCI system research in general through the lenses of three theoretical frameworks that reflect on materiality, i.e. new materialism, indigenous epistemologies, and post-colonial theory on skill mastery. I believe this analysis can offer alternative perspectives that promote a more inclusive understanding of the relationship between humans, materials, and technology in the context of digital fabrication.

CCS CONCEPTS

• **Human-centered computing** → **Interactive systems and tools.**

KEYWORDS

Craft, Art, Digital Fabrication, CAM, Computational System

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1 INTRODUCTION

My research objective is to develop domain-specific action-oriented digital fabrication systems that integrate knowledge and material expertise from craft practices. Expert craftspeople engage with materiality through the design and fabrication of intricately customized and visually appealing artifacts often in the context of low-volume manufacturing. Digital fabrication systems research shares similar objectives with the craft production process and can learn from these manual workflows. In my dissertation, I seek to integrate manual craft knowledge in digital fabrication workflows by supporting low-level craft-specific interactions between humans, materials, and machines.

Craft, argues Glenn Adamson, is not merely a classification of objects, institutions, or people but “only exists in motion ... [as] a way of doing things [1].” Craft workflows require artists to engage with materials through iterative actions, for instance, hand-coiling a clay pot layer by layer, chip-carving a pattern in a piece of wood

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through repeated movements of a knife, or creating an embroidery design one stitch at a time with a needle. These dynamic processes involve continuous feedback loops where the material adapts to the forces at play, and the craftsperson to the material’s responses [19]. I define these manual workflows as *action-oriented*: form results from each action taking place between the craftsperson and the material [4].

Action-oriented workflows, however, extend beyond manual craft and can foster *digital craft-machine-ship*, the shared autonomy between a craftsperson and a machine [2]. To achieve this shared autonomy, digital workflows can engage with materials in a manner similar to manual workflows, i.e., through the execution of low-level actions. These actions can be operationalized using computational abstractions that describe human-machine-material interactions occurring at the material level. Prior HCI research has highlighted that effective abstractions can be informed by domain-expert embodied and technical knowledge, emphasizing the importance of collaborations between HCI researchers and craftspeople [8]. I theorize therefore that action-oriented systems, informed by domain-expert knowledge, have the potential to:

- (1) Facilitate the integration of manual and computational workflows by operating through the same set of actions.
- (2) Enable the execution of complex design tasks that are challenging to accomplish using conventional digital fabrication pipelines.
- (3) Leverage the unique properties of materials to create forms and textures by offering precise control over machine toolpath specifications.

In my thesis work, I present three projects that explore the intersection of craft, digital fabrication, and system design. First, I developed *CoilCAM*, an action-oriented parametric system for clay 3D printing that enables the mathematical specification of machine toolpath [4]. Second, with my colleague, Mert Toka, I created *CeramWrap*, a ceramic ornamentation system that adapts to non-linear manual-computational workflows [17]. Third, I am currently working on *millipath*, a system drawing from woodworking practices that supports the design of surface textures and their fabrication with CNC milling machines [Submission In Progress]. Finally, to expand the theoretical landscape of digital fabrication research in HCI, I intend to examine these contributions through the lens of three frameworks that provide broader perspectives on the dynamics between materials, humans, and fabrication technologies : (1) new materialism, (2) indigenous epistemologies, and (3) post-colonial theories on mastery of skills and matter. This dissertation work is summarized in Figure 1.

2 COILCAM: TRACING PATHS WITH CLAY

Hand-coiling pottery is the practice of rolling thin cylinders of clay and layering them in succession to create three-dimensional pots. The path traced by the craftsperson’s hands determines the final form. Clay 3D printers employ a similar fabrication method, extruding coils of clay and compressing them layer by layer. However, available clay printing-specific systems (software like Clayon [13] or Potterware [9]) use abstractions that fundamentally differ from manual hand-coiling. These strategies follow the “canonical workflow of digital fabrication [18],” where a craftsperson is expected to create a 3D model of the artifact using CAD-based design software,

automatically generate a machine toolpath using a CAM-based tool (e.g. a slicer), and send the resulting operations to the machine for fabrication. This workflow is often material-agnostic and does not provide fine control over human-machine-material actions typical to craft practices.

HCI researchers [6, 16] and artists [5, 12] using additive fabrication technology have developed alternatives to this canonical workflow that better align with manual making processes. These alternatives have shown that granular control over machine toolpath specification, often using CAM-based tools for design tasks rather than CAD-based ones, allows for greater expressivity of material qualities.

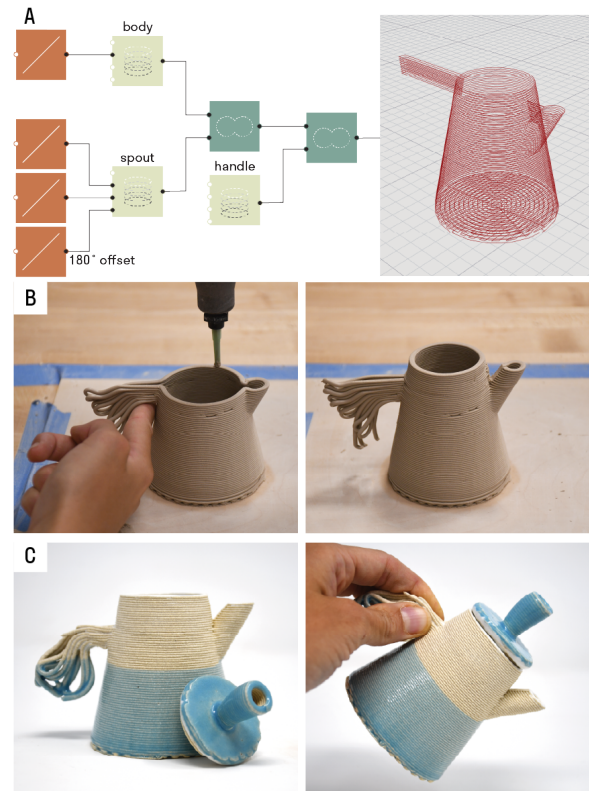


Figure 2: CoilCAM. A) The *CoilCAM* visual programming system is based on toolpath unit generators that can be modified with mathematical functions and Boolean operators to generate machine toolpaths. B) Low-level action-oriented system can enable a craftsperson to plan for human-machine collaboration. C) Teapot with a gravity-shaped human-supported handle made with *CoilCAM*.

Based on these insights, I developed *CoilCAM*, a clay printing-specific visual programming system that enables craftspeople to mathematically specify machine toolpaths [4]. *CoilCAM* was created in the context of an artist residency, which took place in the Expressive Computation Lab at UCSB during the Summer of 2022. It was developed in collaboration with two professional ceramicists, Pilar Wiley, and Avi Farber. Their insights into manual ceramics fabrication and their explorations with clay 3D printers have informed

CoilCAM's abstractions, which include a toolpath unit generator, mathematical shaping functions, and Boolean operators as shown in Figure 2A.

An evaluation through the recreation of handmade pieces by Pilar and Avi and a collaborative discussion following this evaluation revealed critical considerations of action-oriented systems for clay:

- (1) Rapid and iterative variations enabled by a parametrizable visual programming system mirror some aspects of how craftspeople work with material through continuous feedback.
- (2) Continuous and discrete mathematical functions, when mapped to formal components of a pot, e.g., the radial shape of a layer, the profile of a vessel, and the relative rotation and translation between layers, provide opportunities to harness material properties for creating expressive forms and surface textures.
- (3) Low-level machine action representations can support the planning for human-machine-material collaboration (see Figure 2B).

These findings indicate that action-oriented workflows present exciting possibilities for digital fabrication system research.

3 CERAMWRAP: EMBRACING NON-LINEAR WORKFLOWS

Ceramic ornamentation is concerned with decorating clay vessels using various techniques, which encompass additive and subtractive methods. While each technique has its unique approach to creating surface patterns, they all share the challenging condition of adapting to the ever-changing nature of clay. Throughout the process of transforming wet clay into solid ceramics, which involves different drying and firing stages, the material undergoes shrinkage and distortion in often unexpected ways. Craftspeople must therefore adjust their designs and methods to accommodate these dynamic states. As a result, the workflows in ceramic ornamentation often follow non-linear procedures and are difficult to simulate accurately.

Based on the analysis of a ceramics ornamentation design space [17], I observed that these ornamentations are typically composed of decorative elements organized across grid structures with varying levels of alignment. This alignment ranges from *grid-free*, where the elements are arranged on the surface with no discernible persistence in positioning, to *grid-based*, where the positioning of the elements follows a mathematically-defined structure. Furthermore, some decoration techniques, such as underglaze brushing and sgraffito carving, work through the manual application of each decorative element on the surface, which provides flexible and shape-specific alignment to the vessel geometry but can become tedious or challenging to reproduce consistently. In contrast, other techniques, such as underglaze transfer or pattern rollers on slabs, are limited in how the pattern can conform to the specificity of a vessel's geometry but can be applied systematically on large sections of the vessel at a time.

To support the dynamic qualities of clay and the different spatial organization levels of decorative elements, Mert Toka and I developed *CeramWrap*, an adaptable workflow that integrates manual

and digital procedures for the decoration of ceramic vessels. *CeramWrap*'s computational tools support pattern design on radially-symmetrical 3D models of physical vessels, which can be unrolled into templates and fabricated using digital fabrication machines like laser or vinyl cutters. Manual decorations are then applied using these vessel-conforming templates. *CeramWrap*'s non-linear workflow is illustrated in Figure 3.

Demonstrative examples created with *CeramWrap* revealed key aspects of manual-computational approaches to ceramic surface decoration.

- (1) Digital fabrication procedures and existing manual ceramics craft techniques can be integrated into the same workflow: The demonstrative artifacts were decorated using a range of techniques (underglaze brushing, drill carving, sgraffito) and fabricated with different methods (slab-building, wheel-throwing, clay 3D printing).
- (2) Manual-computational workflows can adapt to the evolving nature of materials: The *CeramWrap* workflow enabled rapid changes in the design of patterned templates in order to accommodate the shrinkage and distortion of the ceramic vessels.
- (3) Manual-computational workflows can support different levels of technical requirements: *CeramWrap* was used by craftspeople showing various degrees of skill at different steps of the fabrication process. It also accommodated varying tolerance in aligning templates to vessels, depending on the specific ornamentation techniques employed.

These findings demonstrate the opportunities of digital fabrication systems in supporting non-linear workflows.

4 MILLIPATH: TRACING PATHS THROUGH WOOD

Surface decoration plays a crucial role in various woodworking applications, such as customizing furniture, crafting textured wall panels, and sculpting intricate pieces. Typically, these decorative techniques require the craftspeople to execute a series of carving actions, following a grid-like structure across the surface of the artifact. I see the potential of a digital fabrication system for surface decoration in woodworking that follows a similar workflow by combining the action-oriented machine toolpath specification of *CoilCAM* with the manual-computational surface patterning workflow of *CeramWrap*. In addition, because HCI research has primarily focused on additive fabrication when investigating action-oriented workflows, I am interested in the possibilities of applying this approach to a subtractive fabrication system. To explore these ideas further, I am developing a system called *millipath*, which focuses on the design and fabrication of surface patterns using CNC milling machines. With *millipath*, I seek to establish a set of abstractions that enable craftspeople to specify low-level carving actions while using computational strategies comparable to those found in *CeramWrap* to organize these actions on a surface in space.

Initial investigations with *millipath* recall David Pye's notion of risk associated with manual craft [14]. Pye associated machine processes to the *workmanship of certainty*. *millipath*, however, exemplifies how intentionally choosing to control certain features

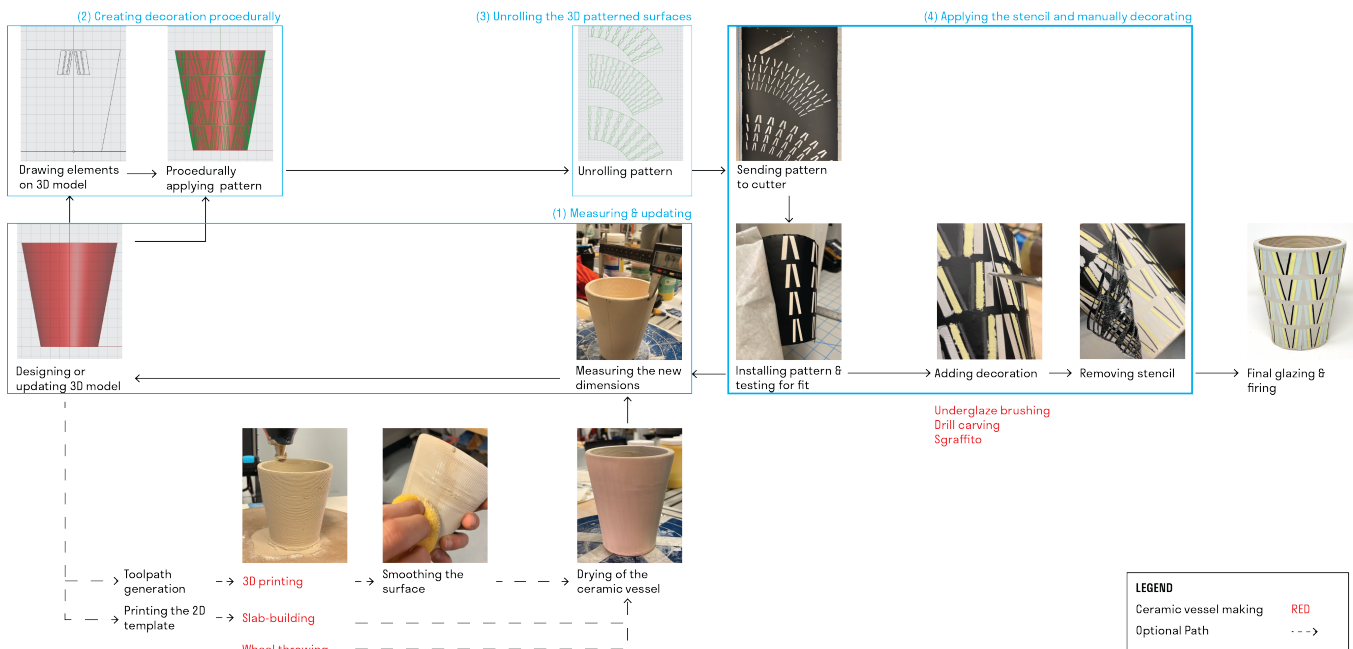


Figure 3: *CeramWrap* is a non-linear workflow that integrates existing manual fabrication and decoration techniques (in red). The flowchart shows a simplified version of the workflow and it's four different stages (in blue). The images represent the production steps of a cup decorated with chevron elements brushed with underglaze.

of the process (toolpath), while relinquishing others (milled material simulation), enables craftspeople to encounter uncertainty, surprises, and failures inherent to the *workmanship of risk*.

5 FUTURE WORK

While the projects presented above explore digital fabrication approaches from the perspective of machine actions, collaboration with domain-expert, and artifacts productions, I believe that analyzing these contributions through theoretical frameworks that critically examine the relationship between humans, materials, and technology can yield further insights for HCI system research. In my future dissertation work, I intend to focus on three related yet distinct frameworks: (1) new materialism, (2) indigenous epistemologies, and (3) post-colonial theory.

Being and becoming: New Materialism is a cross-disciplinary philosophical current that conceives agency as distributed among an assemblage of *things* [3]. Through this conceptual framework, matter – living and non-living indistinctly – is considered active and always evolving rather than passive and changeless [20]. Donna Haraway employs the term *sympoiesis* – making-with – to describe the intricate relationships that exist within “complex, dynamic, responsive, situated, historical systems [10].”

Origins and forms of knowledge: In *Indigenous Methodologies*, Margaret Kovach presents a perspective on indigenous knowledge that encompasses multiple sources that are not solely cognitive but also embodied, instinctual, and spiritual. Knowledge transcends human boundaries to animals, birds, fish, insects, the earth, land, sky, and spirit. Knowledge is also rooted in the collective and interdependent relationships that individuals share with the places

they inhabit. It is, therefore, non-anthropocentric but perceived as a reciprocal exchange that must be maintained between all parties involved in order to maintain “an ecological and cosmological balance [11].”

Mastery and control: Engaging in post-colonial theory, Julietta Singh suggests that “mastery invariably and relentlessly reaches toward the indiscriminate control over something – whether human or inhuman, animate or inanimate [15].” Even when mastery is used in the context of expertise, it continues to reproduce an imbalance of power through hierarchical structures. These ideas ground Ingold’s rejection of the hylomorphic model of making further into the socio-political causes and consequences of control over matter. To resist the subordination of nature and people, Deane Curtin suggests adopting an ethic of care [7], which echoes David Pye’s definition of craft as a form of care for the material in use [14].

By employing these conceptual frameworks, I aim to explore the notion of craft-making as an interconnected ecosystem and investigate the implications of Haraway's concept of *making-with* for the participating and non-participating (but impacted) agents in the making process. Furthermore, I am intrigued by the idea of developing manual-computational systems rooted in care. I intend to examine how such systems align or diverge from dualistic and hierarchical perspectives on making with materiality.

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REFERENCES

- [1] Glenn Adamson. 2007. *Thinking Through Craft*. Berg Publishers.
- [2] Kristina Andersen, Ron Wakkary, Laura Devendorf, and Alex McLean. 2019. Digital crafts-machine-ship: creative collaborations with machines. *Interactions* 27, 1 (Dec. 2019), 30–35. <https://doi.org/10.1145/3373644>
- [3] Jane Bennett. 2010. *Vibrant Matter: A Political Ecology of Things*. Duke University Press.
- [4] Samuelle Bourgault, Pilar Wiley, Avi Farber, and Jennifer Jacobs. 2023. CoilCAM: Enabling Parametric Design for Clay 3D Printing Through an Action-Oriented Toolpath Programming System. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23)*. Association for Computing Machinery, New York, NY, USA, 1–16. <https://doi.org/10.1145/3544548.3580745>
- [5] Sarah Rosalena Brady. 2020. transposing a form. <https://www.sarahrosalena.com/transposing-a-form>
- [6] Leah Buechley and Franklin Pezutti-Dyer. 2022. Extruder Turtle. <https://handandmachine.org/index.php/2020/10/23/extruder-turtle/>
- [7] Deane Curtin. 1991. Toward an Ecological Ethic of Care. *Hypatia* 6, 1 (1991), 60–74. <https://www.jstor.org/stable/3810033> Publisher: [Hypatia, Inc., Wiley].
- [8] Laura Devendorf, Katya Arquilla, Sandra Wirtanen, Allison Anderson, and Steven Frost. 2020. Craftspeople as Technical Collaborators: Lessons Learned through an Experimental Weaving Residency. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (CHI '20)*. Association for Computing Machinery, New York, NY, USA, 1–13. <https://doi.org/10.1145/3313831.3376820>
- [9] Emerging Objects. 2020. POTTERWARE. <https://www.potterware.com/>
- [10] Donna J. Haraway. 2016. *Staying with the Trouble: Making Kin in the Chthulucene*. Duke University Press. Google-Books-ID: ZvDgDAAAQBAJ.
- [11] Margaret Kovach. 2010. *Indigenous Methodologies: Characteristics, Conversations, and Contexts*. University of Toronto Press. Google-Books-ID: TqyVJ2xc6E0C.
- [12] LIA. 2013. Filament Sculptures. <https://www.liaworks.com/theprojects/filament-sculptures/>
- [13] MicroPlus Technologies, Inc. 2022. Clayon. <https://www.clayontec.com/en/>
- [14] David Pye. 1971. *The nature and art of workmanship*. London, Studio Vista; New York, Van Nostrand Reinhold. <http://archive.org/details/natureartofwor00pyed>
- [15] Julietta Singh. 2017. *Unthinking Mastery: Dehumanism and Decolonial Entanglements*. Duke University Press. Google-Books-ID: hiFBDwAAQBAJ.
- [16] Blair Subbaraman and Nadya Peek. 2022. p5.fab: Direct Control of Digital Fabrication Machines from a Creative Coding Environment. In *Designing Interactive Systems Conference (DIS '22)*. Association for Computing Machinery, New York, NY, USA, 1148–1161. <https://doi.org/10.1145/3532106.3533496>
- [17] Mert Toka, Samuelle Bourgault, Camila Friedman-Gerlicz, and Jennifer Jacobs. 2023. An Adaptable Workflow for Manual-Computational Ceramic Surface Ornamentation. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software & Technology (UIST '23)*. Association for Computing Machinery, San Francisco, California, USA.
- [18] Hannah Twigg-Smith, Jasper Tran O'Leary, and Nadya Peek. 2021. Tools, Tricks, and Hacks: Exploring Novel Digital Fabrication Workflows on #PlotterTwitter. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21)*. Association for Computing Machinery, New York, NY, USA, 1–15. <https://doi.org/10.1145/3411764.3445653>
- [19] Norbert Wiener. 2010. What is Cybernetics? In *The Craft Reader*, Glenn Adamson (Ed.). Berg Publishers.
- [20] Susanne Witzgall. 2017. Power of Material/ Politics of Materiality an Introduction. In *Power of Material/Politics of Materiality*. Diaphanes, 13–26. Google-Books-ID: 6AEVMQAACAAJ.